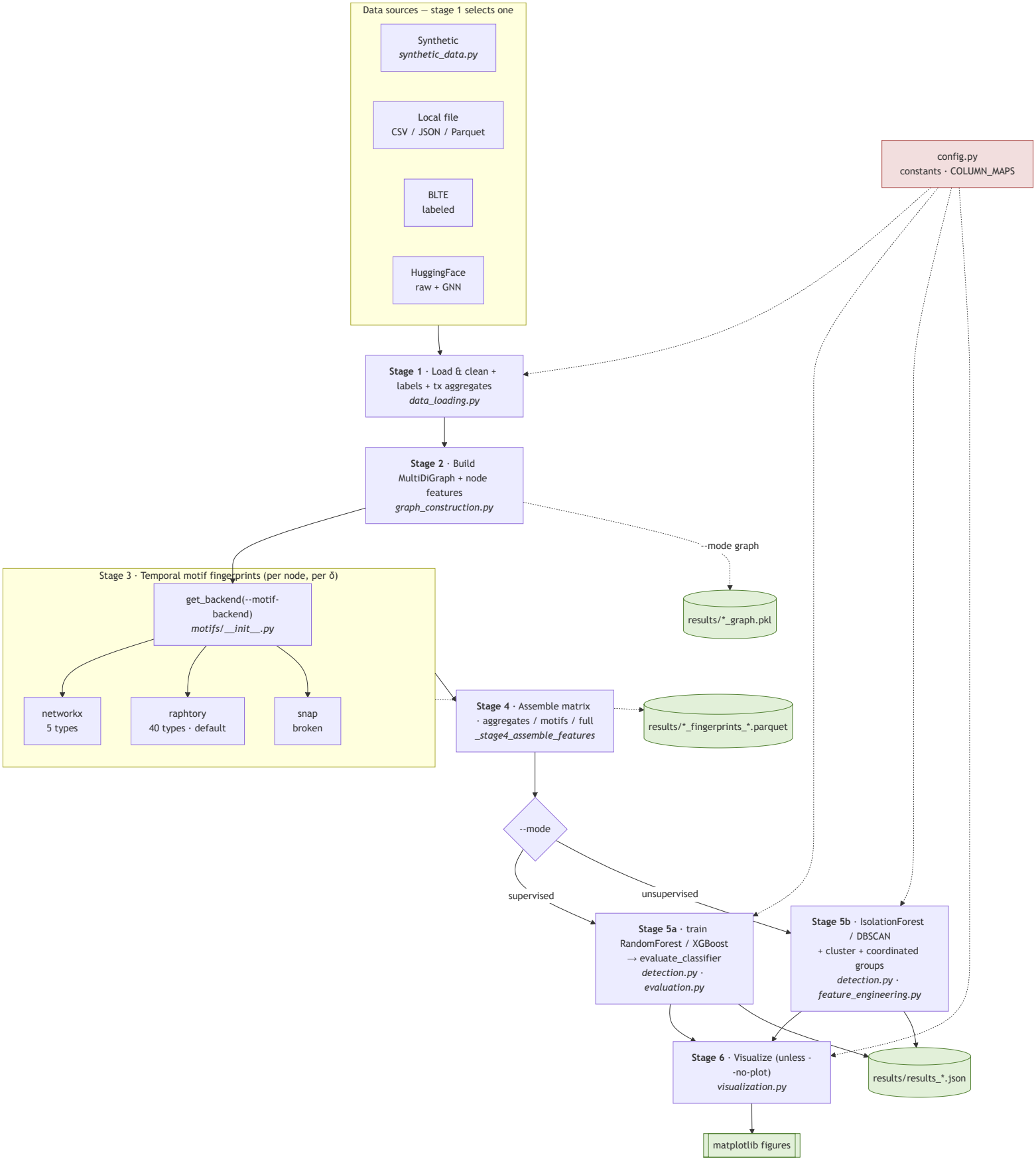

Data sources — stage 1 selects one
Synthetic
synthetic_data.py
Local file
CSV / JSON / Parquet
BLTE
labeled
HuggingFace
raw + GNN
config.py
constants · COLUMN_MAPS
Stage 1 · Load & clean + labels + tx aggregates
data_loading.py
Stage 2 · Build MultiDiGraph + node features
graph_construction.py
--mode graph
results/*_graph.pkl
Stage 3 · Temporal motif fingerprints (per node, per δ)
get_backend(--motif-backend)
motifs/__init__.py
networkx
5 types
raphtory
40 types · default
snap
broken
Stage 4 · Assemble matrix · aggregates / motifs / full
_stage4_assemble_features
results/*_fingerprints_*.parquet
--mode
supervised
unsupervised
Stage 5a · train RandomForest / XGBoost → evaluate_classifier
detection.py · evaluation.py
Stage 5b · IsolationForest / DBSCAN + cluster + coordinated groups
detection.py · feature_engineering.py
Stage 6 · Visualize (unless --no-plot)
visualization.py
results/results_*.json
matplotlib figures